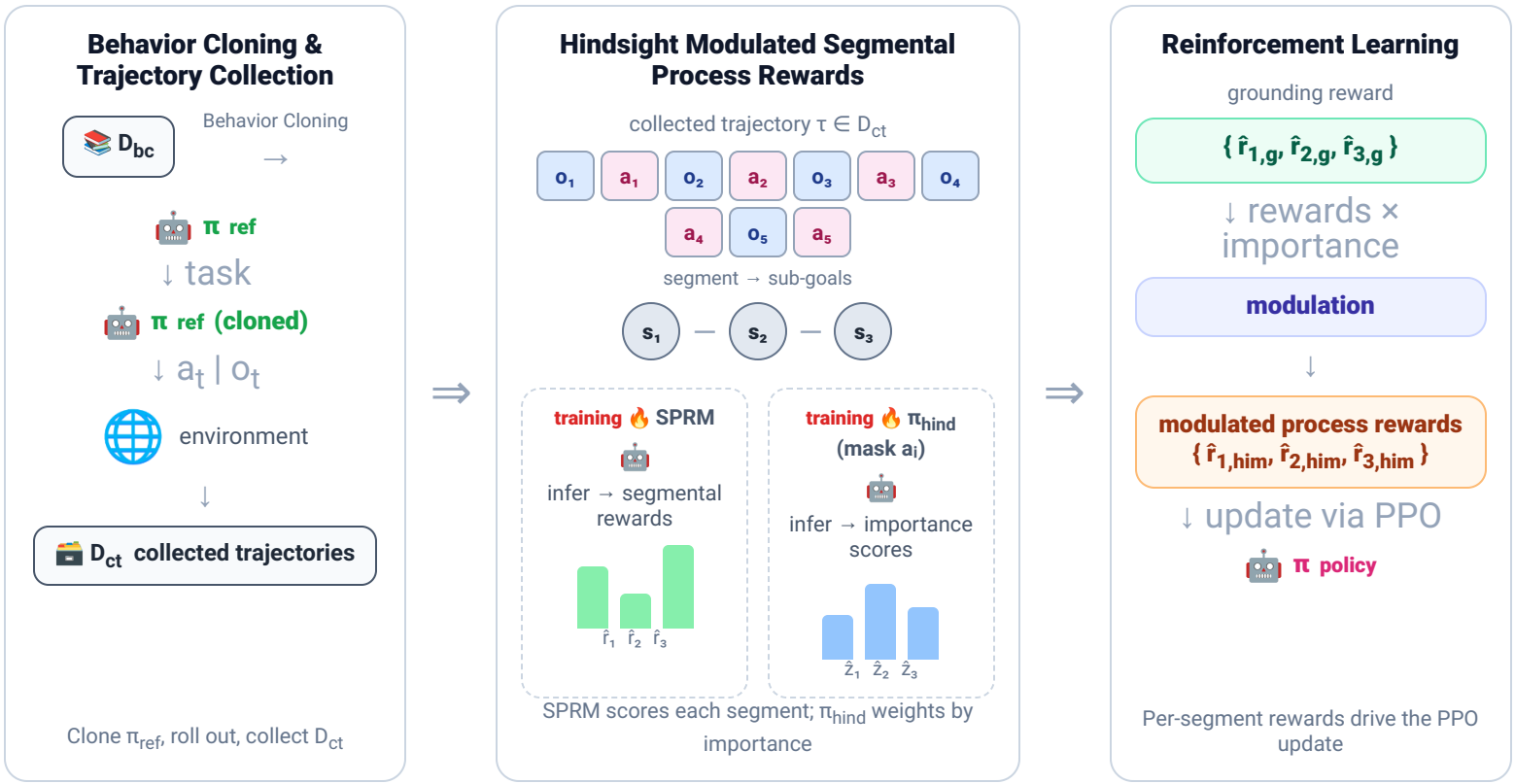


SeRPO: Segment-level Rubric Process Rewards for AppWorld Agents



Why segment-level process rewards matter

**LEARNS WHERE OUTCOME RL CAN'T**

On **fail-only** trajectories every rollout shares the same outcome, so outcome-reward baselines (Vanilla GRPO, GiGPO) get equal rewards  $\rightarrow$  group variance 0  $\rightarrow$  **gradient = 0** (no learning). SeRPO's **per-segment** rewards still vary within a trajectory  $\rightarrow$  **gradient  $\neq 0$** , so the policy keeps improving even when all trajectories failed.

**LOCATES THE FAILURE POINT**

A single outcome scalar says only *pass / fail*. Per-segment rewards  $\{\hat{r}_1, \hat{r}_2, \hat{r}_3\}$  attribute credit along the trajectory, so a low-scoring segment **pinpoints which step caused the failure** (e.g. a premature action with missing API exploration) — turning an opaque rollout into a diagnosable, interpretable signal.

**Vanilla GRPO / GiGPO**  
outcome only

$\times, \times, \times, \times$

all equal  $\rightarrow \Delta=0$   
 $\times$  no gradient

**SeRPO**  
per segment

5, 4, 1, 4

scores vary  $\rightarrow \Delta \neq 0$   
 $\checkmark$  still learns

$\downarrow$  credit assigned to each segment  $\rightarrow$  PPO policy update  $\downarrow$

**Outcome-RL**  
one scalar

., ., ., .

flat / 0  $\rightarrow$  policy unchanged

**SeRPO**  
per segment

$\blacktriangle, \blacktriangle, \blacktriangledown, \blacktriangle$

reinforce good  $\cdot$  suppress failure step

**advantage  $\hat{A}_i$**   
from segment reward

$+, +, -, +$

PPO:  $\uparrow \pi$  if  $\hat{A} > 0$ ,  $\downarrow \pi$  if  $\hat{A} < 0$

**effect on policy**  
 $\pi(\text{action} | \text{state})$

$\pi \uparrow, \pi \uparrow, \pi \downarrow, \pi \uparrow$

good steps made more likely,  
failure step less likely

$\pi$  policy updated — failure segment's action down-weighted

PPO uses each segment's advantage  $\hat{A}_i$ : it **increases** the probability of actions in  $+\hat{A}$  segments and **decreases** it for the  $-\hat{A}$  failure step. Outcome-RL's flat/zero advantage gives no such directed push.